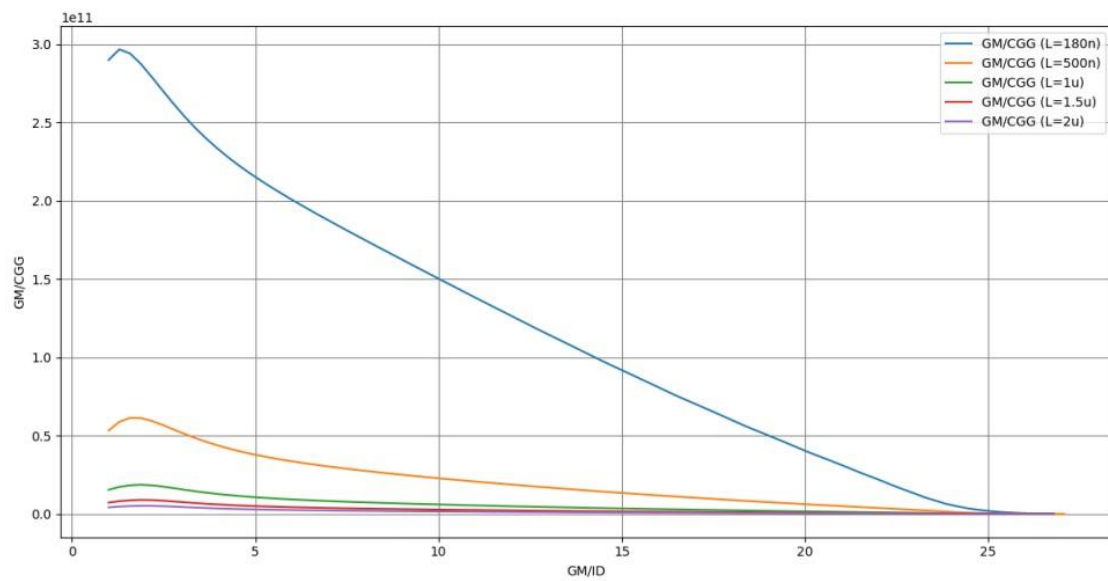
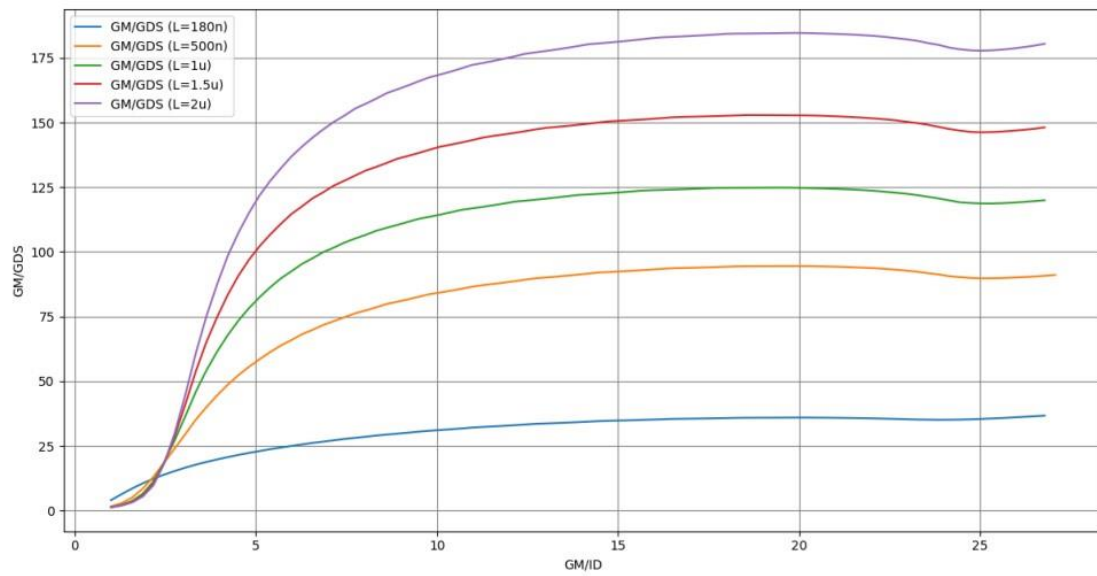
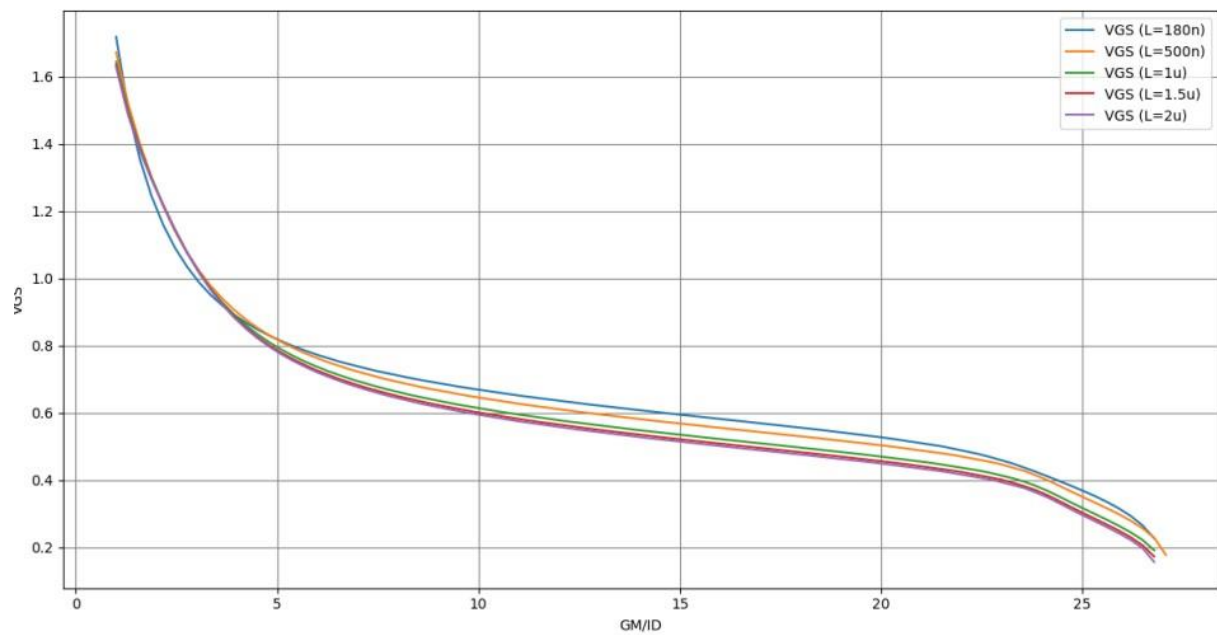
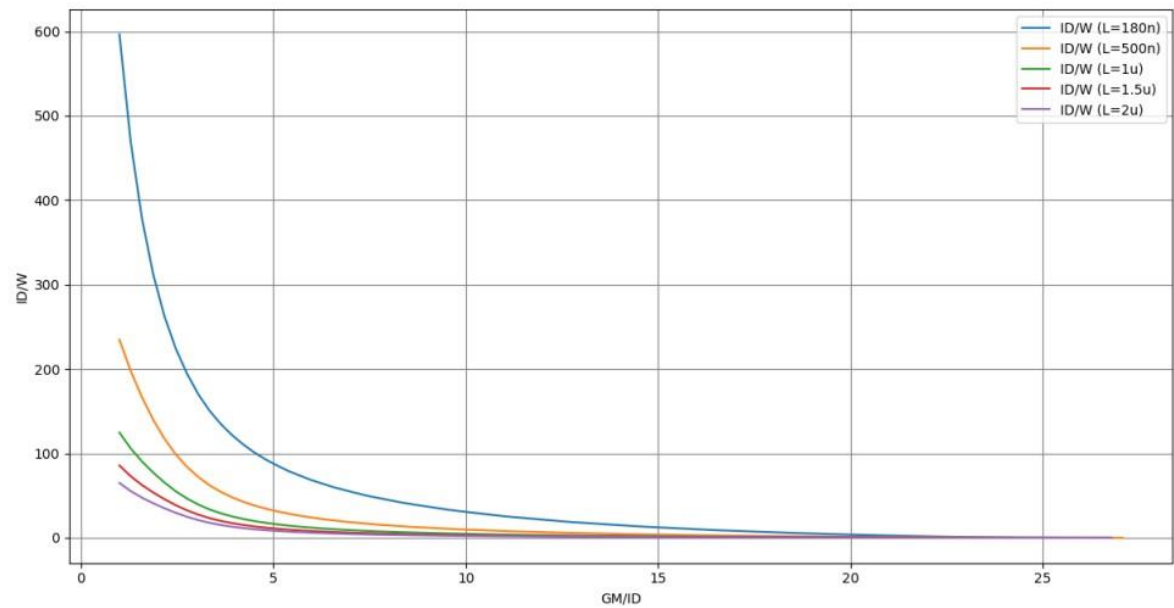


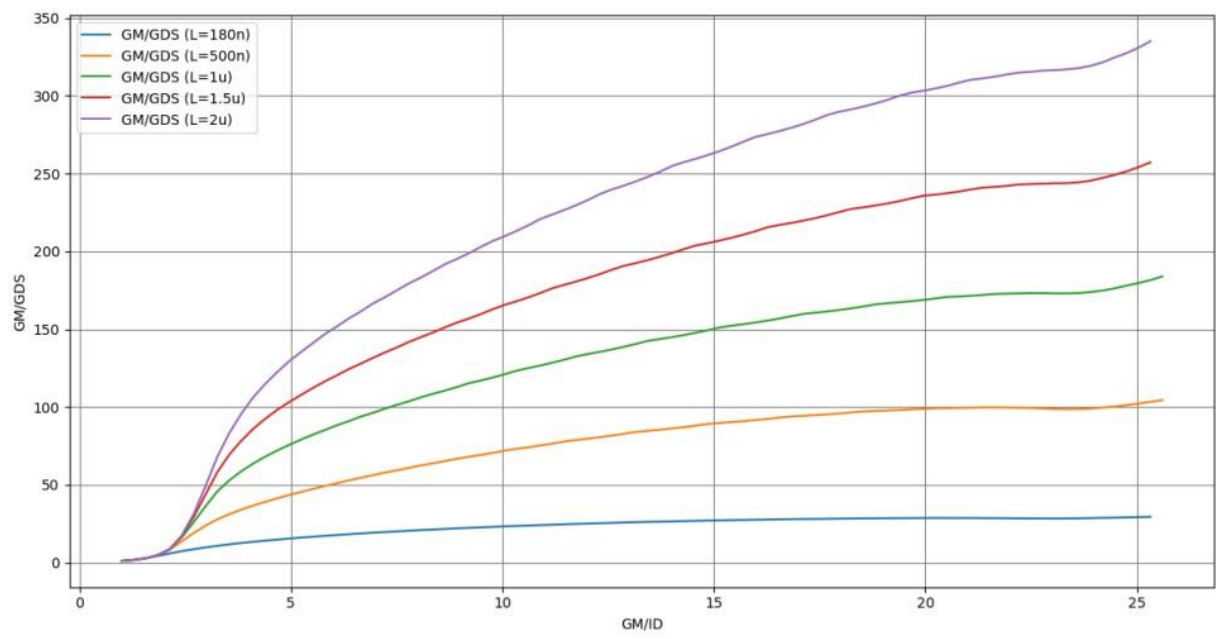
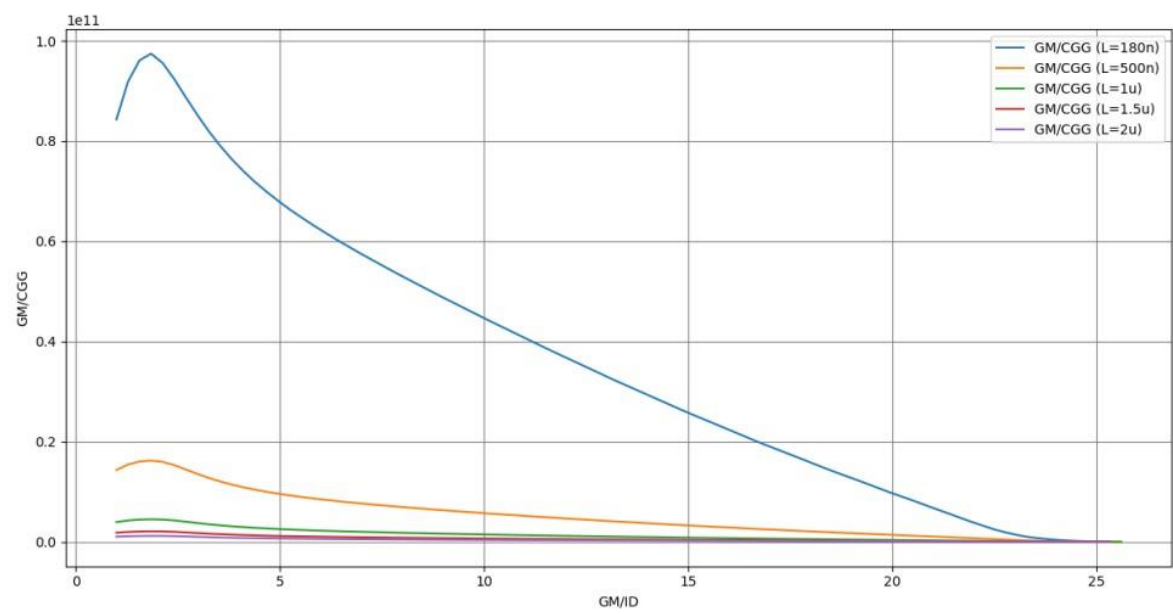
Part 1: GM/ID design charts

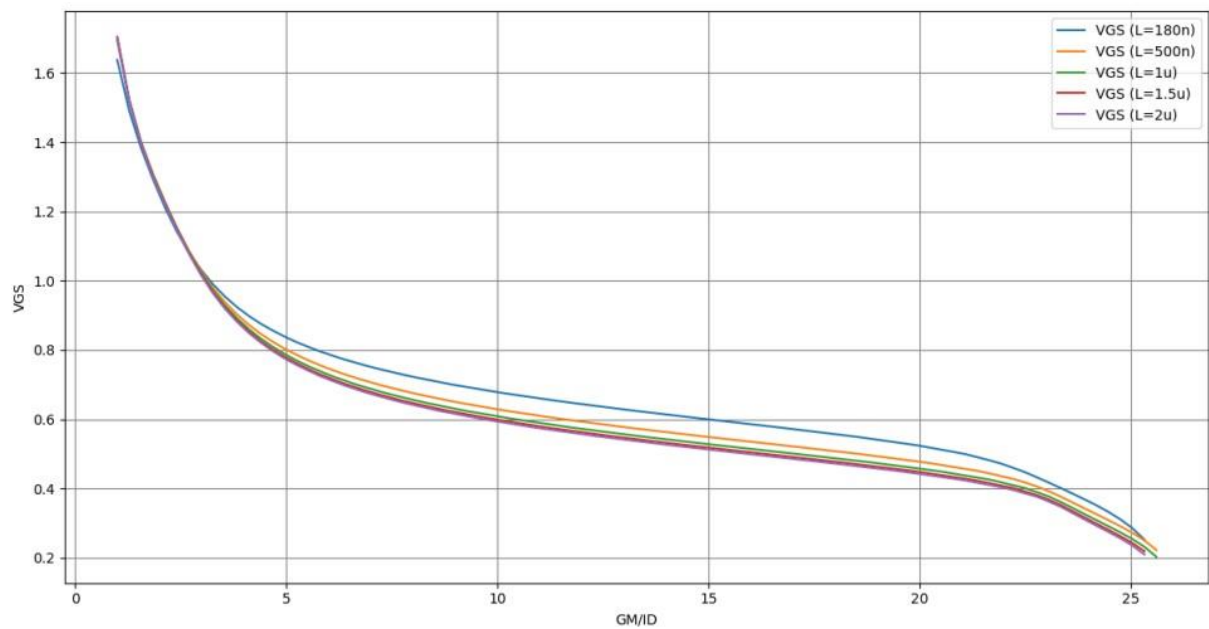
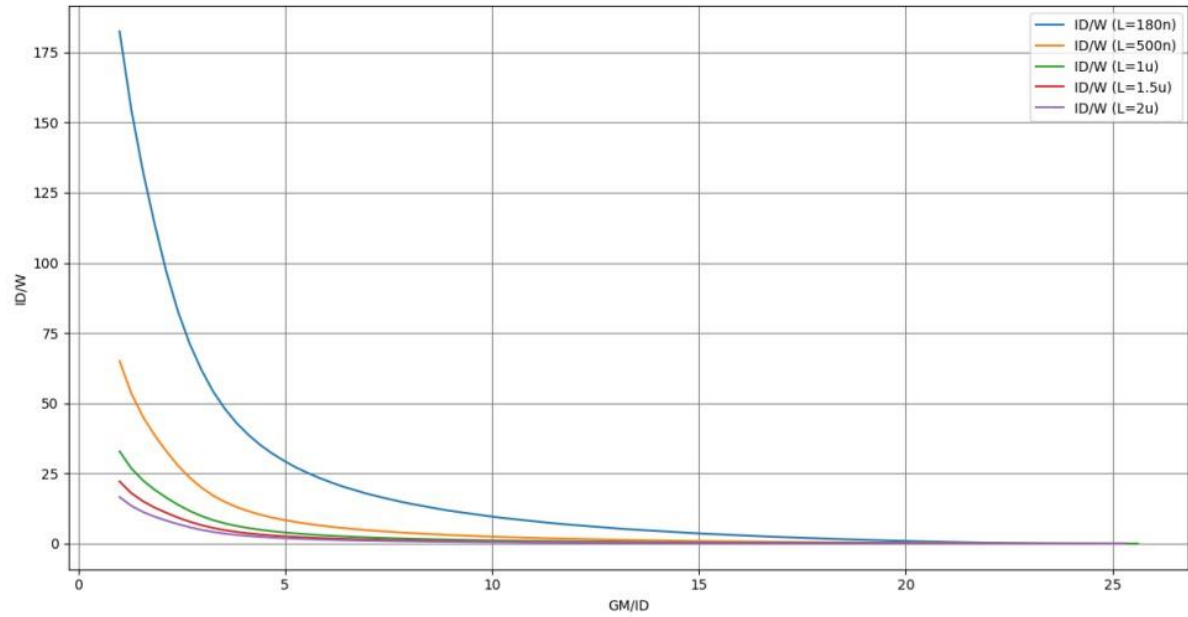
NMOS_CHARTS





PMOS CHART





PART 2: OTA Design

FIRST TRY SA

Note the following : we will follow Doctor Hesham suggested sizing procedure then reiterate to achieve the specs

Specs DataSheet

OTA Current $\rightarrow 80 \mu A$

CMIR-low = $-V_{th_{in}} + V_{DS_{CMN}}$ or $-V_{th_{in}} - V_{GS_{CASN}} + V_{CASN}$

CMIR-high = $-V_{GS_{in}} - V_{CS}^* + V_{DD}$

Closed loop gain $\approx \frac{1}{\beta} = 2 \rightarrow \beta = \frac{1}{2}$

loop gain = PAo.L $\rightarrow 60dB \rightarrow A_{o.L} \approx 2000$

Closed loop bandwidth = $10MHz = BW_{o.L} * (1 + LG) \rightarrow BW_{o.L} \approx 10kHz$

NOTE THAT THIS DATASHEET IS BY NO MEANS AN ACCURATE REPRESENTATION OF THE RESULTS BUT WILL TRY TO USE IT TO HELP GUIDE THE DESIGN PROCEDURE

;

We will use OTA current of $80 \mu A$ and split the current between the input branch and the cascode branch equally by $20 \mu A$ each also the current Mirror NMOS transistor will carry double that current ($40 \mu A$)

Following the DOCTOR suggested assumptions we have :

M_input

LUT

Corner

Temp (°C)

State1

ID

gm/ID

L

VDS

VSB

Stack

Y-Expr

Device Parameters

#	Parameter	Value
1	ID	20u
2	L	200n
3	W	6.3u
4	VGS	589.3m
5	VDS	675m
6	VSB	0
7	gm/ID	14.68
8	Vstar	136.3m
9	ft	3.401G
10	gm/gds	34.34
11	VA	2.339
12	ID/W	3.175
13	gm/W	46.6
14	AREA	1.26p
15	gm	293.6u
16	gmb	86.85u
17	gds	8.549u
18	ro	117k
19	VTH	479.1m
20	VDSAT	105.7m

M: hiToggleMagnifier()

M_CS(above input)

LUT

Corner

Temp (°C)

State1

ID

gm/ID

L

VDS

VSB

Stack

Y-Expr

Device Parameters

#	Parameter	Value
1	ID	40u
2	L	1u
3	W	34.55u
4	VGS	609.6m
5	VDS	675m
6	VSB	0
7	gm/ID	9.863
8	Vstar	202.8m
9	ft	237.6M
10	gm/gds	123.3
11	VA	12.5
12	ID/W	1.158
13	gm/W	11.42
14	AREA	34.55p
15	gm	394.5u
16	gmb	122.4u
17	gds	3.2u
18	ro	312.5k
19	VTH	409.6m
20	VDSAT	160.9m

M: hiToggleMagnifier()

log: /... Library Manager: Director

M_CASN

LUT

nch

Corner

tt

Temp (°C)

27.0

State1

Save State

ID

20u

gm/ID

15

L

0.5u

VDS

0.45

VSB

0

Stack

1

Get

Apply

Y-Expr

ID/CGG

Plot

Replace

Append

Device Parameters

#	Parameter	Value
1	ID	20u
2	L	500n
3	W	5.31u
4	VGS	570.7m
5	VDS	450m
6	VSB	0
7	gm/ID	14.78
8	Vstar	135.3m
9	ft	2.152G
10	gm/gds	84.7
11	VA	5.731
12	ID/W	3.766
13	gm/W	55.67
14	AREA	2.655p
15	gm	295.6u
16	gmb	80.46u
17	gds	3.49u
18	ro	286.5k
19	VTH	442.6m
20	VDSAT	105.6m

M: hiToggleMagnifier()

g: /...

Library Manager: Director

M_CASP

ne/user01/projects/ex_LUTs/

LUT

pch

Corner

tt

Temp (°C)

27.0

State1

Save State

ID

20u

gm/ID

15

L

0.5u

VDS

0.45

VSB

0

Stack

1

Get

Apply

Y-Expr

ID/CGG

Plot

Replace

Append

Device Parameters

#	Parameter	Value
1	ID	20u
2	L	500n
3	W	21.73u
4	VGS	549.7m
5	VDS	450m
6	VSB	0
7	gm/ID	14.98
8	Vstar	133.5m
9	ft	513.7M
10	gm/gds	82.16
11	VA	5.485
12	ID/W	920.4m
13	gm/W	13.79
14	AREA	10.86p
15	gm	299.6u
16	gmb	90.74u
17	gds	3.647u
18	ro	274.2k
19	VTH	432.9m
20	VDSAT	105.6m

M_CMN

ne/user01/projects/ex_LUTs/

LUT: nch

Corner: tt

Temp (°C): 27.0

State1: Save State

ID: 40u

gm/ID: 10

L: 1u

VDS: 0.45

VSB: Set VDS or VDB.

Stack: 1

Get Apply

Y-Expr: ID/CGG

Plot Replace Append

Device Parameters

#	Parameter	Value
1	ID	40u
2	L	1u
3	W	8.59u
4	VGS	616m
5	VDS	450m
6	VSB	0
7	gm/ID	9.856
8	Vstar	202.9m
9	ft	960.6M
10	gm/gds	99.5
11	VA	10.09
12	ID/W	4.657
13	gm/W	45.9
14	AREA	8.59p
15	gm	394.3u
16	gmb	107.5u
17	gds	3.962u
18	ro	252.4k
19	VTH	406.2m
20	VDSAT	157.8m

M: hiToggleMagnifier()

g: /... Library Manager: Directo

M_CMP

ne/user01/projects/ex_LUTs/

LUT: pch

Corner: tt

Temp (°C): 27.0

State1: Save State

ID: 20u

gm/ID: 10

L: 1u

VDS: 0.45

VSB: Set VDS or VDB.

Stack: 1

Get Apply

Y-Expr: ID/CGG

Plot Replace Append

Device Parameters

#	Parameter	Value
1	ID	20u
2	L	1u
3	W	17.47u
4	VGS	610.4m
5	VDS	450m
6	VSB	0
7	gm/ID	9.857
8	Vstar	202.9m
9	ft	234.7M
10	gm/gds	108.4
11	VA	11.28
12	ID/W	1.145
13	gm/W	11.28
14	AREA	17.47p
15	gm	197.1u
16	gmb	61.17u
17	gds	1.818u
18	ro	550.1k
19	VTH	409.6m
20	VDSAT	161.5m

M: hiToggleMagnifier()

From previous results if we assume V_{star} of all transistor will be around 0.2

$$VICM - low = -V_{th_{inupt}} - V_{GS_{CASN}} + V_{CAS_N}$$

From specs $VICM - low = 0 \rightarrow$ we get that V_{CAS_N} can be as high as 1V

$$V_{CAS_p} = -V_{GS_{VCASP}} - V_{CMP}^* + V_{DD}$$

V_{CAS_p} can go as high as 1.05V but we will choose it to be around 0.9

$$V_{CAS_N} = 1 \quad \text{and} \quad V_{CAS_p} = 0.9$$

To find VREF we need to find VoMAX and VoMIN

$$Vo_{min} = -Vth_{CASN} + VCAS_N = 0.56$$

$$Vo_{max} = Vth_{CASP} + VCAS_P = 1.34$$

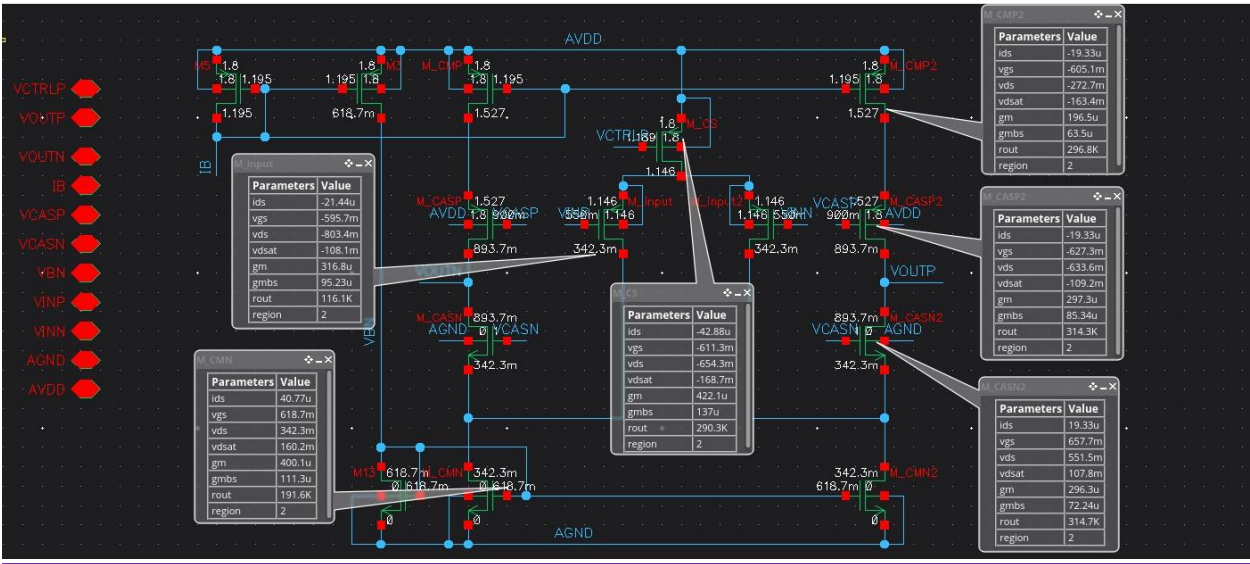
to maximize the swing $V_{OCM} = 0.95$ and we will choose it to be 0.9

$$\mathbf{VREF = 0.9}$$

PART 3: Open-Loop OTA Simulation (Behavioral CMFB)

We will now start simulating and after seeing the results we can go back to the DATA-SHEET and trade-off matrix to reiterate

IMP NOTE : WE WILL SEE THE ANALYSIS RESULTS AND GO BACK TO SOLVE ALL THE QUESTIONS AFTER REDOING THE DESIGN



AC analysis results to see if any reiteration is needed

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
icDesign02_lab_...	dB20(v"/VODIF...				
icDesign02_lab_...	ymax(dB20(v"/V...	66.68			
icDesign02_lab_...	bandwidth(v"/V...	21.73K			
icDesign02_lab_...	gainBwProd(v"/V...	47.01M			
icDesign02_lab_...	unityGainFreq(v...	47.32M			
icDesign02_lab_...	phaseMargin(v"/...	87.87			

From the following results if we go back to the DATA-Sheet and trade-off matrix we can see that we probably will need to increase the open loop gain (note that the bandwidth from data sheet is way higher than needed so increasing the gain shouldn't be an issue)

We know that the open loop gain will be equal to

$$A_{O.L} = g_{m_{input}} R_{OUT}$$

$$R_{OUT} = \left[(g_m + g_{mb})_{CASN} * r_{o_{CASN}} * r_{o_{CMN}} // (g_m + g_{mb})_{CASP} * r_{o_{CASP}} * r_{o_{CMP}} // (g_m + g_{mb})_{input} * r_{o_{input}} * r_{o_{CS}} \right]$$

Going back to the balloons to see the bottlenecks of the circuit.

$$R_{OUT} = [13.7M // 35M // 14.5M]$$

The bottleneck on R_{OUT} is coming mainly from the input branch and CASN branch and if we look closely from : rout (input and NMOS current mirrors) this makes sense since we have chosen short channel length for the input and the NMOS current mirror carries double the current

Reiteration:

I will decrease the current consumption from 80uA to 70uA to increase rout of the NMOS current mirror the 70uA will be divided as follows: 20u A to input stage (same as before)

35uA to the NMOS current mirrors and 15uA to the cascode branch(note that this change will change rout for CASN and CASP and PMOS current mirror but we will decrease the fallen VDS on all of them and also slightly decrease their L) ; we will also try to distribute the supply voltage differently in an attempt to increase rout of the input and NMOS CM , we will also slightly increase L of the input

NOTE: I know that changing VDS in the SA to increase rout might not mean much since VDS is usually decided by the VGS and BIAS voltages in the circuit but going back to the DATA-sheet we can manipulate the bias voltages while keeping in mind the input ranges and output swing requirements

All this gives the following :

M_input

M_CS(above input)

LUTs Directory

ne/user01/projects/ex_LUTs/

LUTpch

Cornertt

Temp (°C)27.0

State1

Save State

ID20u

gm/ID15

L400n

VDS0.7

VSB0

Stack1

Get

Apply

Y-ExprID/CGG

Plot

Replace

Append

Device Parameters

#	Parameter	Value
1	ID	20u
2	L	400n
3	W	16.1u
4	VGS	556.3m
5	VDS	700m
6	VSB	0
7	gm/ID	14.76
8	Vstar	135.5m
9	rT	829.1M
10	gm/gds	78.53
11	VA	5.321
12	ID/W	1.242
13	gm/W	18.33
14	AREA	6.44p
15	gm	295.2u
16	gmb	88.77u
17	gds	3.758u
18	ro	266.1k
19	VTH	440.7m
20	VDSAT	106m

LUTs Directory

ne/user01/projects/ex_LUTs/

LUTpch

Cornertt

Temp (°C)27.0

State1

Save State

ID40u

gm/ID10

L1u

VDS0.4

VSB0

Stack1

Get

Apply

Y-ExprID/CGG

Plot

Replace

Append

Device Parameters

#	Parameter	Value
1	ID	40u
2	L	1u
3	W	35.07u
4	VGS	610.5m
5	VDS	400m
6	VSB	0
7	gm/ID	9.856
8	Vstar	202.9m
9	rT	233.7M
10	gm/gds	100.9
11	VA	10.23
12	ID/W	1.141
13	gm/W	11.24
14	AREA	35.07p
15	gm	394.3u
16	gmb	122.3u
17	gds	3.909u
18	ro	255.8k
19	VTH	409.6m
20	VDSAT	161.6m

M_CASN

ne/user01/projects/ex_LUTs/ ...

LUTnch

Corrertt

Temp (°C)27.0

State1 Save State

ID15u

gm/ID15

L0.5u

VDS0.36

VSB0

Stack1

Get Apply

Y-ExprID/CGG

Plot Replace Append

Device Parameters

#	Parameter	Value
1	ID	15u
2	L	500n
3	W	4.01u
4	VGS	571.2m
5	VDS	360m
6	VSB	0
7	gm/ID	14.8
8	Vstar	135.1m
9	fT	2.134G
10	gm/gds	75.84
11	VA	5.125
12	ID/W	3.741
13	gm/W	55.36
14	AREA	2.005p
15	gm	222u
16	gmb	60.44u
17	gds	2.927u
18	ro	341.7k
19	VTH	443.3m
20	VDSAT	105.5m

ICADVM20.1-64b - Log: ... Library

M_CASP

ne/user01/projects/ex_LUTs/ ...

LUTpch

Corrertt

Temp (°C)27.0

State1 Save State

ID15u

gm/ID15

L0.5u

VDS0.36

VSB0

Stack1

Get Apply

Y-ExprID/CGG

Plot Replace Append

Device Parameters

#	Parameter	Value
1	ID	15u
2	L	500n
3	W	16.49u
4	VGS	549.9m
5	VDS	360m
6	VSB	0
7	gm/ID	15
8	Vstar	133.4m
9	fT	507M
10	gm/gds	75.08
11	VA	5.006
12	ID/W	909.6m
13	gm/W	13.64
14	AREA	8.245p
15	gm	225u
16	gmb	68.12u
17	gds	2.996u
18	ro	333.7k
19	VTH	432.9m
20	VDSAT	105.8m

CADVM20.1-64b - Log: ... Library

M_CMN

:/user01/projects/ex_LUTs/ ...

LUT

Corner

Temp (°C)

State1 Save State

ID

gm/ID

L

VDS

VSb

Stack

Get Apply

Y-Expr

Plot Replace Append

Device Parameters

#	Parameter	Value
1	ID	35u
2	L	1u
3	W	7.46u
4	VGS	614.6m
5	VDS	700m
6	VSb	0
7	gm/ID	9.849
8	Vstar	203.1m
9	rT	968.7M
10	gm/gds	118.7
11	VA	12.05
12	ID/W	4.692
13	gm/W	46.21
14	AREA	7.46p
15	gm	344.7u
16	gmb	93.94u
17	gds	The sizing and parameters
18	ro	344.4k
19	VTH	404.7m
20	VDSAT	157.9m
21	cgg	56.64f
22	cgs	50.11f
23	capd	3.377f

Manager: Directory ...e/user0... Virt

M_CMP

ne/user01/projects/ex_LUTs/ ...

LUT

Corner

Temp (°C)

State1 Save State

ID

gm/ID

L

VDS

VSb

Stack

Get Apply

Y-Expr

Plot Replace Append

Device Parameters

#	Parameter	Value
1	ID	15u
2	L	500n
3	W	6.16u
4	VGS	630.5m
5	VDS	360m
6	VSb	0
7	gm/ID	9.905
8	Vstar	201.9m
9	rT	887.8M
10	gm/gds	53.21
11	VA	5.372
12	ID/W	2.435
13	gm/W	24.12
14	AREA	3.08p
15	gm	148.6u
16	gmb	45.38u
17	gds	2.792u
18	ro	358.1k
19	VTH	432.9m
20	VDSAT	162.4m

M:

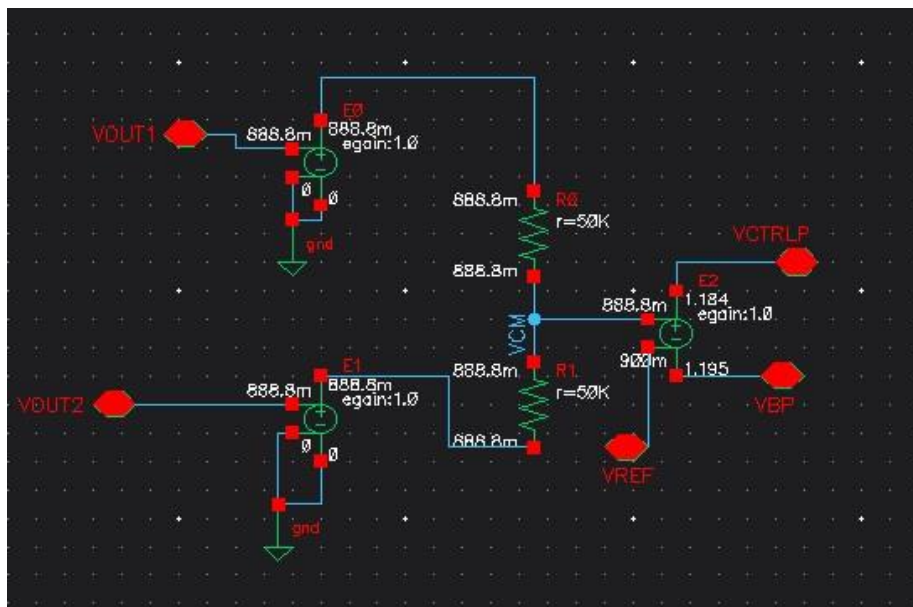
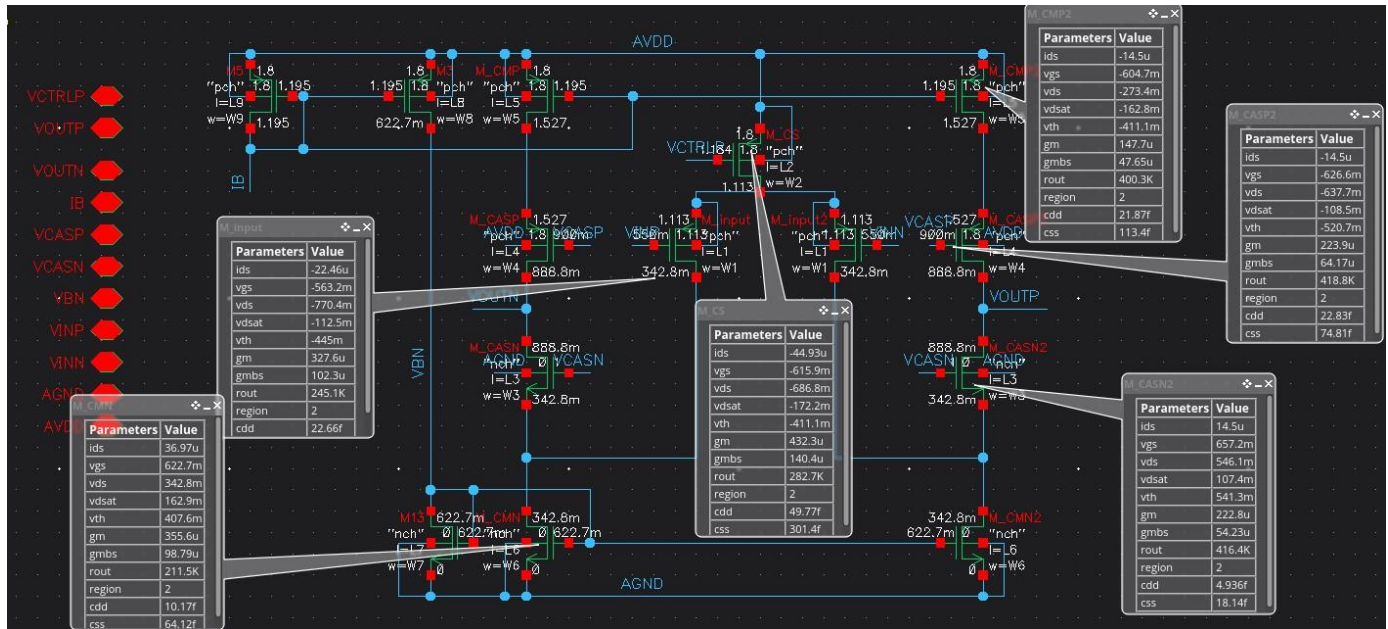
ICADVM20.1-64b - Log: ... Library

Now we go back and redo Part3

PART 3: Open-Loop OTA Simulation (Behavioral CMFB)

Report the following:

1) Schematic of the OTA and bias circuit with DC node voltages and transistors OP parameters (id, vgs, vds, vdsat, vth, gm, gds, region) clearly annotated.



- Set VICM at the middle of the CMIR.
- Select VREF to maximize the symmetrical output swing.

- What is the CM level at the OTA output?

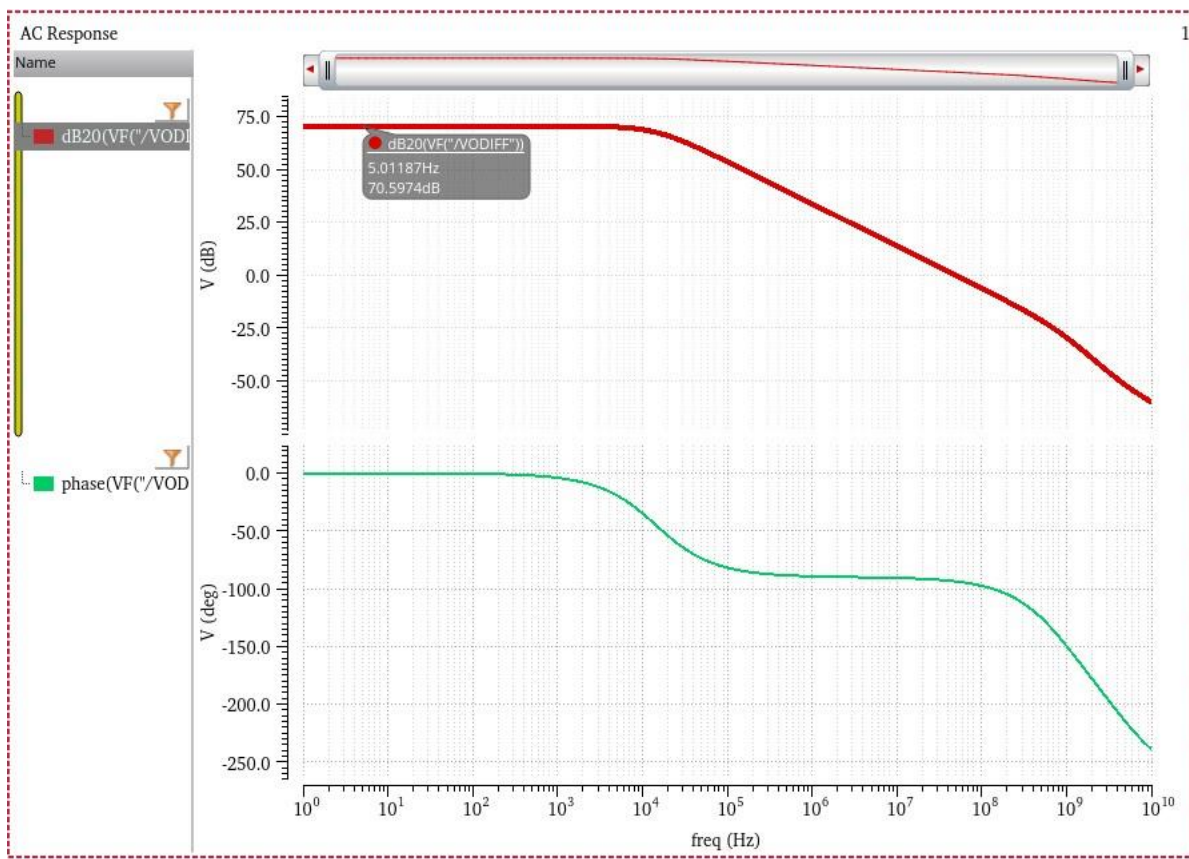
888mV

- What are the differential input and output voltages of the error amplifier? What is the relation between them?

VID(12m V) VOD(12m V), The gain of the error amplifier (1)

2) Diff small signal ccs:

- Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).
- Set VIDAC = 1 and VICMAC = 0.
- Set VICM at the middle of the CMIR.
- Plot diff gain (magnitude in dB and phase) vs frequency.



- Calculate circuit parameters (DC gain, BW, GBW, UGF, and PM).

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
icDesign02_lab_...	dB20(V"/VODIF...				
icDesign02_lab_...	dB20Aol	70.6			
icDesign02_lab_...	Aol	3.387K			
icDesign02_lab_...	BW	14.51K			
icDesign02_lab_...	GBW	49.27M			
icDesign02_lab_...	UGF	49.26M			
icDesign02_lab_...	PM	86.33			

→ Cadence Hint: Use Cadence calculator expressions to calculate circuit parameters (A_o, A_o in dB, BW,

GBW, UGF). You may use Cadence calculator to create other useful expressions.

• Compare simulation results with hand calculations in a table (use SS parameters from OP simulation in your hand analysis).

Hand analysis:

$$A_{O.L} = g_{m_{input}} R_{OUT}$$

$$R_{OUT} = \left[(g_m + g_{mb})_{CASN} * r_{o_{CASN}} * r_{o_{CMN}} // (g_m + g_{mb})_{CASP} * r_{o_{CASP}} * r_{o_{CMP}} // (g_m + g_{mb})_{input} * r_{o_{input}} * r_{o_{CS}} \right]$$

$$BW = \frac{1}{2\pi * R_{out} * (CL + 2C_{dd})}$$

$$GBW = \frac{g_{m_{input}}}{2\pi(CL + 2C_{dd})}$$

$$PM = 90 - \tan^{-1}\left(\frac{w_u}{w_{p2}}\right)$$

We can make an educated guess that second pole will be at the drain of the CS(above input) transistor (this guess is based upon that r_o of this transistor is the highest also it has the highest C_{dd} of all transistors)

$$w_{p2} = \frac{1}{2\pi * \left(\frac{1}{2((g_m + g_{mb})_{input})} \right) * (C_{dd} + 2C_{ss})}$$

	CAD	Hand analysis
Aol	70.6db	70.74db
BW	14.51K	14.85K
GBW	49.27M	50.5M
UGF	49.26M	50.5M
PM	86	Type equation here.

PART 4: Open-Loop OTA Simulation (Actual CMFB)

- The 40uA current is divided between the four CMFB branches. You may assume $L = 1\mu\text{m}$ and $gm/ID = 15$ for all transistors with unknown L or gm/ID for simplicity (note that L and gm/ID for some transistors are already known, why?).

We will this suggestion from the doctor ,I will add a buffer branch and divide the current into 8uA to each of the 5 branches , we do indeed know gm/ID or V_{GS} of some of the transistors example M_{VCTRLP} , M_{VBN} , M_{VBP} their L and V_{GS} will be the same as thoses in last part to avoid the circuit not working properly

This will give

M_{VCTRLP}

M_{VBN}

ADT Sizing Assistant

Settings Help

LUT Settings

LUTs Directory:

LUT:

Corner:

Temp (°C):

State1 Save State

ID:

VGS:

L:

VDS:

VSB:

Stack:

Get Apply

Y-Expr:

Plot Replace Append

Device Parameters

#	Parameter	Value
3	W	6.44u
4	VGS	616m
5	VDS	600m
6	VSB	0
7	gm/ID	9.575
8	Vstar	208.9m
9	rT	243.8M
10	gm/gds	118.1
11	VA	12.33
12	ID/W	1.242
13	gm/W	11.89
14	AREA	6.44p
15	gm	76.58u
16	gmb	23.79u
17	gds	648.6n
18	ro	1.542M
19	VTH	409.6m
20	VDSAT	165.5m
21	cgg	49.99f
22	cgs	43.31f

Virtuoso® Schematic Editor X...

ADT Sizing Assistant

Settings Help

LUT Settings

LUTs Directory:

LUT:

Corner:

Temp (°C):

State1 Save State

ID:

VGS:

L:

VDS:

VSB:

Stack:

Get Apply

Y-Expr:

Plot Replace Append

Device Parameters

#	Parameter	Value
3	W	3.16u
4	VGS	622m
5	VDS	600m
6	VSB	0
7	gm/ID	9.557
8	Vstar	209.3m
9	rT	999.2M
10	gm/gds	113
11	VA	11.82
12	ID/W	5.066
13	gm/W	48.42
14	AREA	3.16p
15	gm	153u

ADT Sizing Assistant

M_VBP

LUTs Directory

Path: `z:/user01/projects/ex_LUTs/`

LUT: `pch`

Corner: `tt`

Temp (°C): `27.0`

State1 `Save State`

ID: `8u`

VGS: `0.605`

L: `500n`

VDS: `0.9`

VSB: `0`

Stack: `1`

`Get` `Apply`

Y-Expr: `ID/CGG`

`Plot` `Replace` `Append`

Device Parameters

#	Parameter	Value
3	W	3.94u
4	VGS	605m
5	VDS	900m
6	VSB	0
7	gm/ID	11.11
8	Vstar	180m
9	rT	826.7M
10	gm/gds	86.59
11	VA	7.791
12	ID/W	2.032
13	gm/W	22.59
14	AREA	1.97p
15	gm	88.99u
16	gmb	27.1u
17	gds	1.028u
18	ro	973k
19	VTH	432.9m
20	VDSAT	143.8m
21	cgg	17.13f
22	cgs	13.85f

Virtuoso® Schematic Editor X...

M_VREF

ADT Sizing Assistant

`Settings` `Help`

▼ LUT Settings

LUTs Directory

Path: `z:/user01/projects/ex_LUTs/`

LUT: `nch`

Corner: `tt`

Temp (°C): `27.0`

State1 `Save State`

ID: `8u`

gm/ID: `15`

L: `1u`

VDS: `0.6`

VSB: `0`

Stack: `1`

`Get` `Apply`

Y-Expr: `ID/CGG`

`Plot` `Replace` `Append`

Device Parameters

#	Parameter	Value
3	W	4.49u
4	VGS	536.8m
5	VDS	600m
6	VSB	0
7	gm/ID	14.66
8	Vstar	136.5m
9	rT	565.8M
10	gm/gds	122.8
11	VA	8.377
12	ID/W	1.782
13	gm/W	26.11
14	AREA	4.49p
15	pm	117.3u

ADT Sizing Assistant

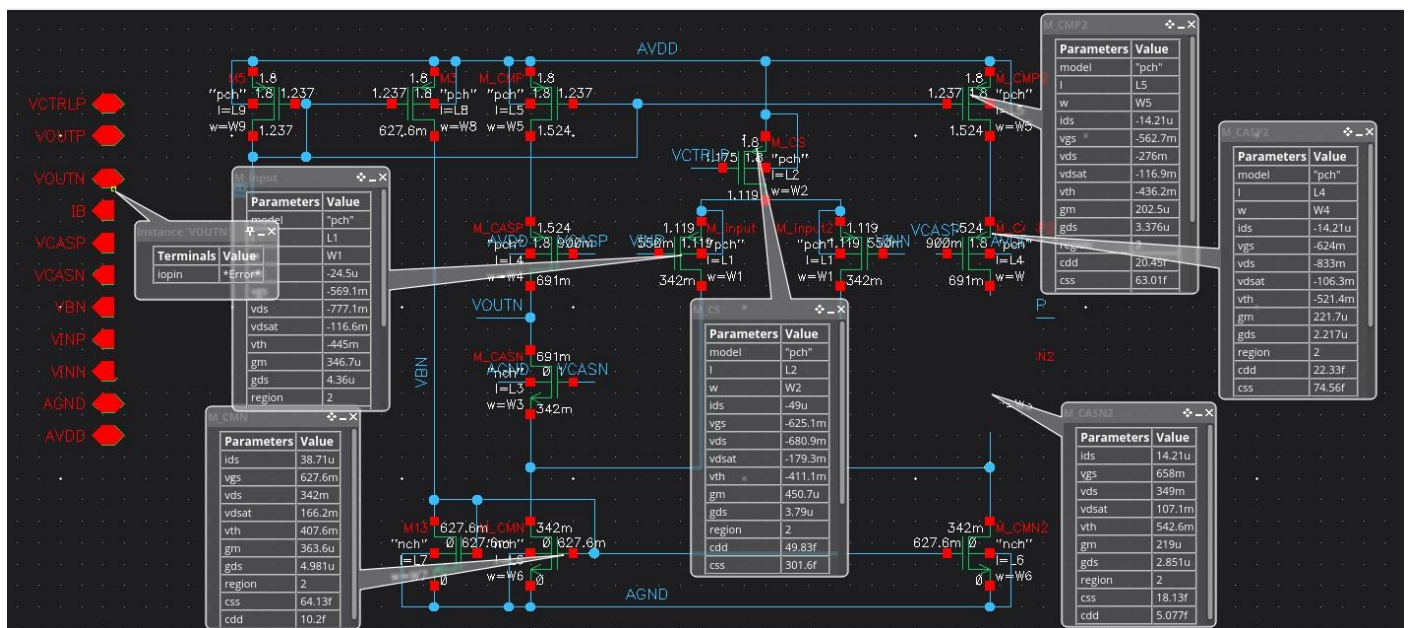
- The output range now is from $2V^*$ to $VDD - V^* - |VGSP|$. Select V_{ocm} to be around the middle of this range to have maximum symmetric output swing (it will be different from the value selected in the behavioral model).

$$V_{o_{min}} = 0.4$$

$$V_{o_{max}} = -V^* - VGS_p + 1.8 = 1.05$$

from this we choose $VREF = 0.7$

1) Schematic of the OTA and CMFB circuit with DC node voltages and transistors OP parameters (id, vgs, vds, vdsat, vth, gm, gds, region) clearly annotated.



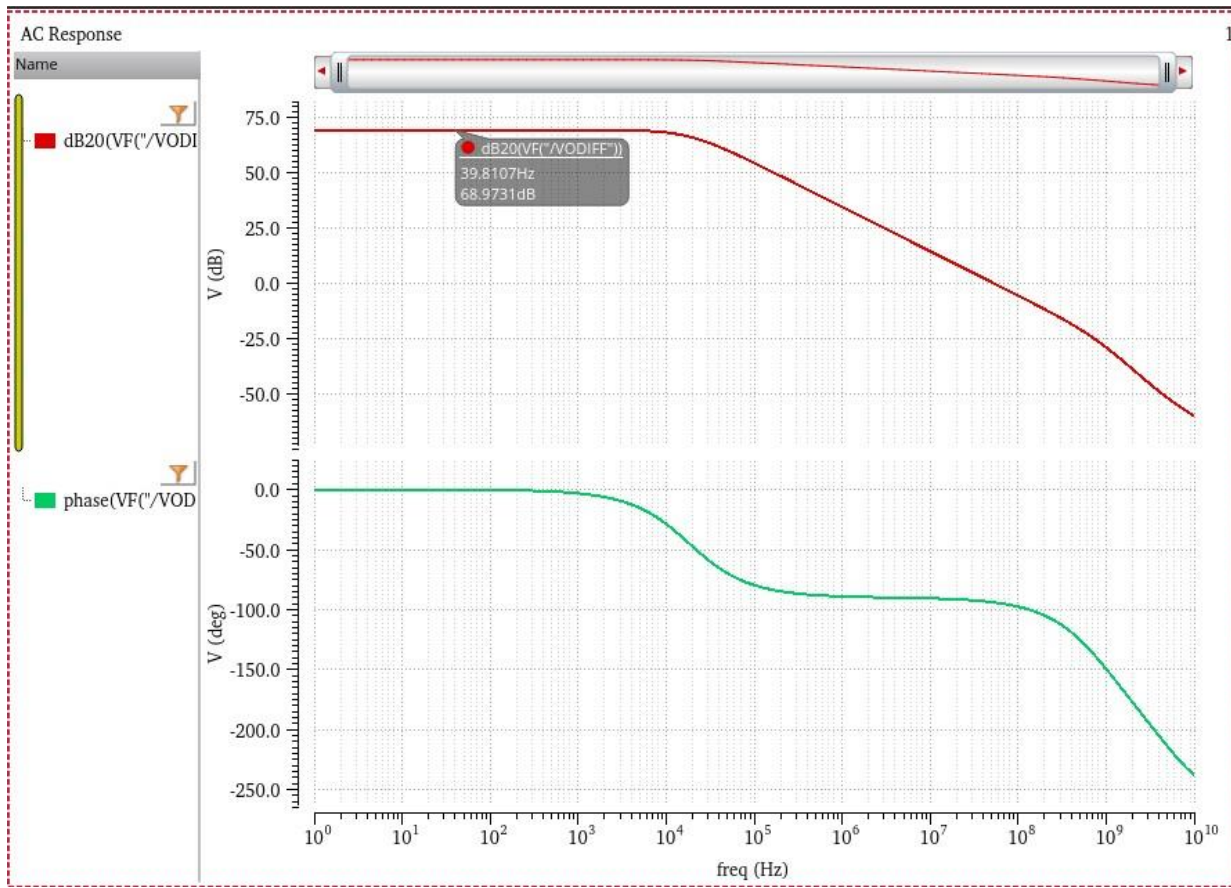
- 691mV, the CMFB aims to adjust V_{OCM} to the desired V_{REF} by controlling V_{CTRLP} and since we have set V_{REF} to 700mV V_{OCM} is very close to that value with a very small variation

- What are the differential input and output voltages of the error amplifier? What is the relation between them?

9mV and 9mV, the gain of the amplifier (close to $g_{m5}/g_{m1} \approx 1$)

2) Diff small signal ccs:

- Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).
- Set VIDAC = 1 and VICMAC = 0.
- Set VICM at the middle of the CMIR.
- Plot diff gain (magnitude in dB and phase) vs frequency.



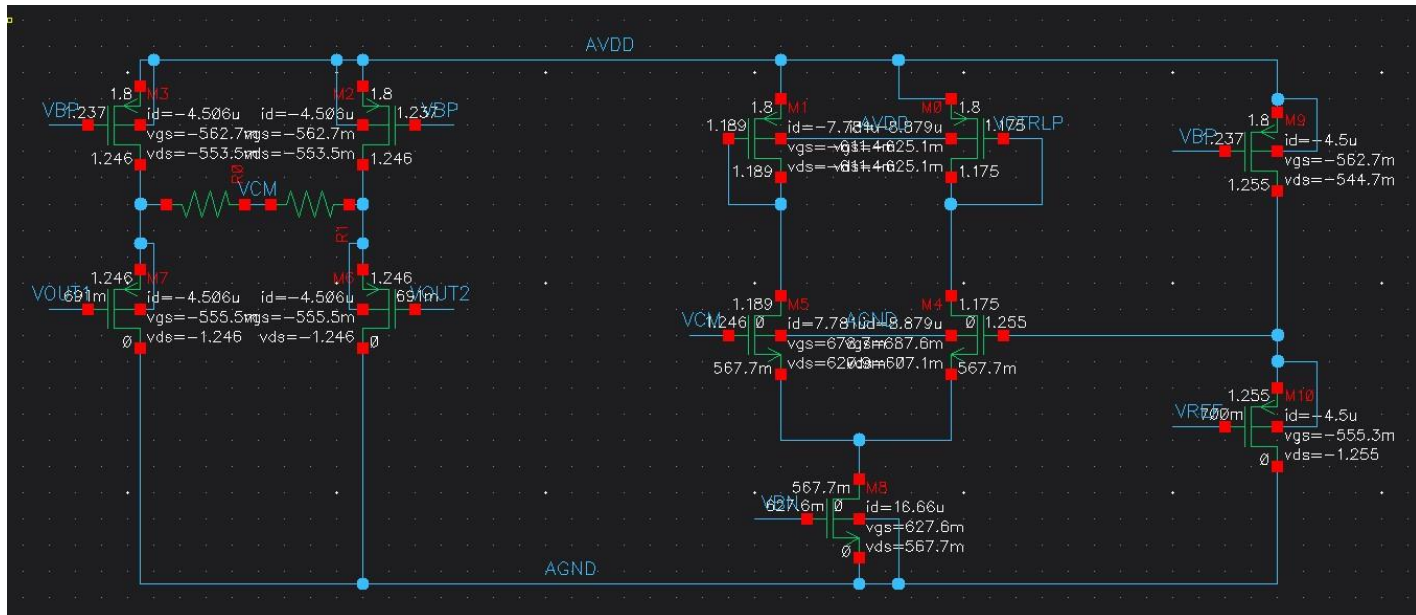
- Calculate circuit parameters (DC gain, BW, GBW, UGF, and PM).

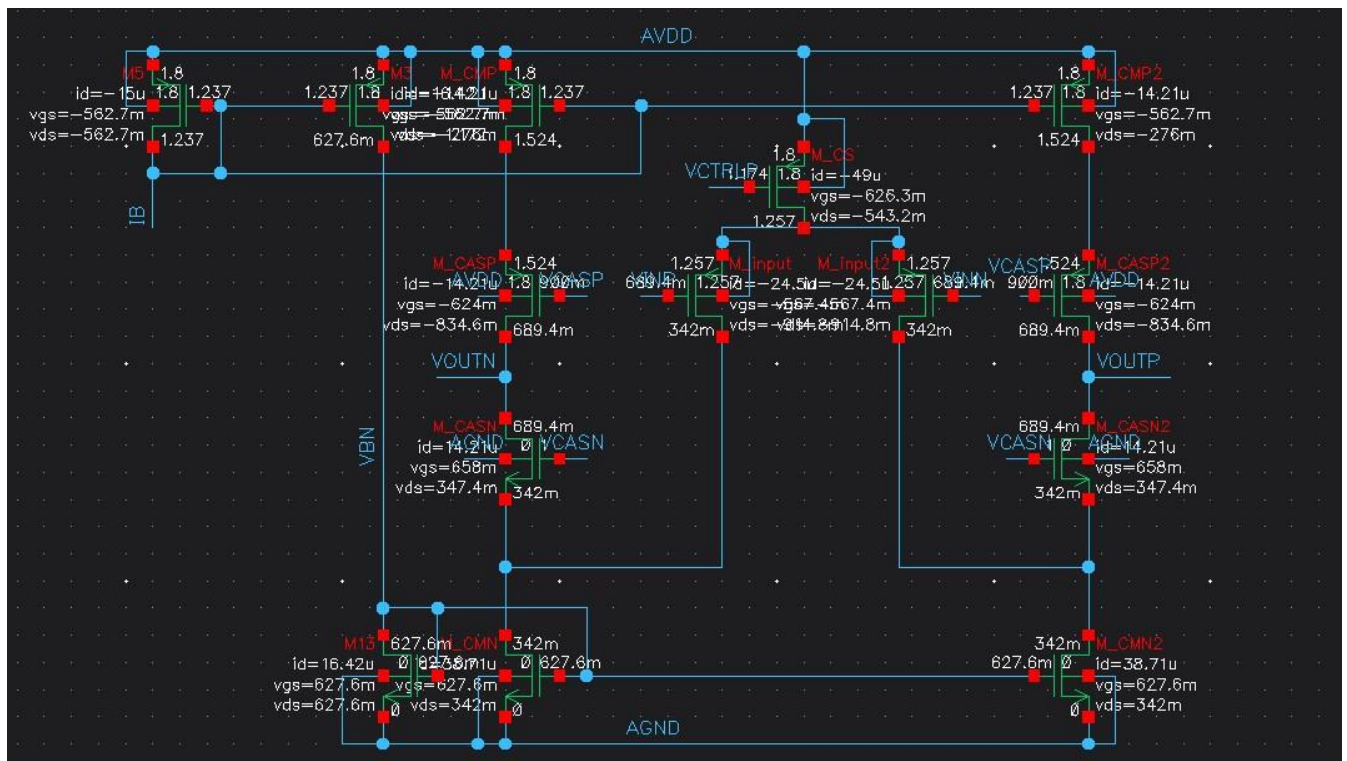
Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
icDesign02_lab_...	dB20(v"/VODIF...				
icDesign02_lab_...	db20Aol	68.97			
icDesign02_lab_...	BW	18.37K			
icDesign02_lab_...	GBW	51.73M			
icDesign02_lab_...	UGF	51.81M			
icDesign02_lab_...	phaseMargin(v"...	86.13			

PART 5: Closed Loop Simulation (AC and STB Analysis)

Report the following:

1) Schematic of the OTA and the CMFB circuit with DC OP point clearly annotated in closed-loop configuration.





- What is the CM level at the OTA output? Why?

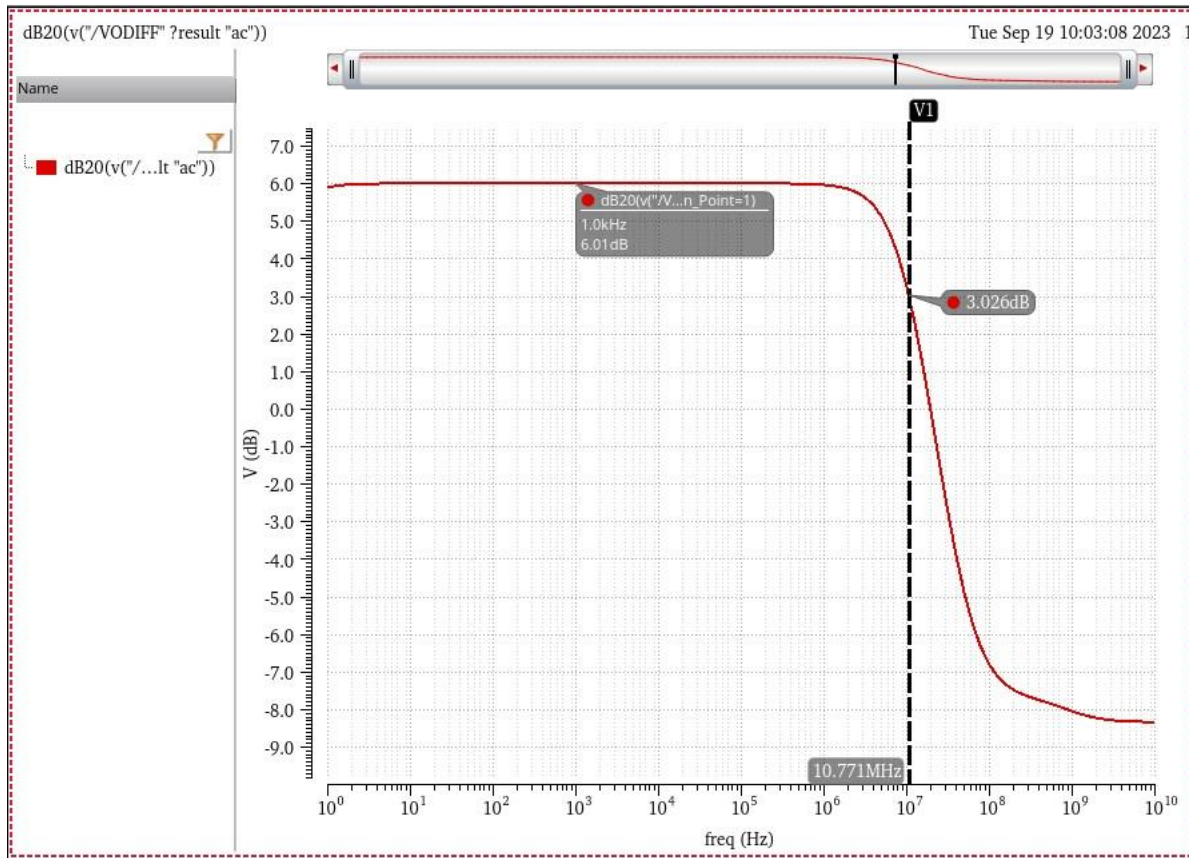
689mV, we set the new VREF to be 700mV this voltage will adjust the output CM level by adjusting VCTRLP

- What is the CM level at the OTA input? Why?

689mV

2) Differential closed-loop response:

- Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).
- Set VIDAC = 1 and VICMAC = 0.
- Plot VODIFF vs frequency
- Use Measures or cursors to calculate circuit parameters (DC gain, CL BW, CL GBW)



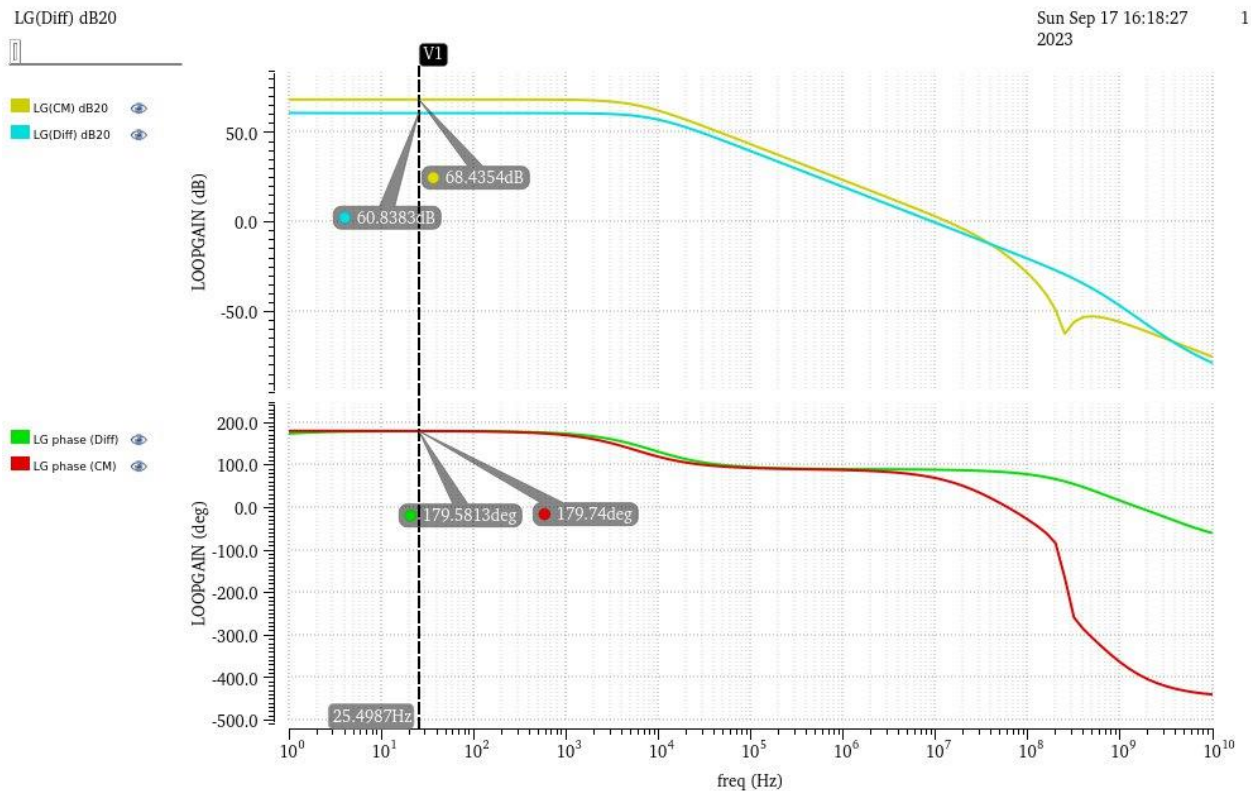
Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
icDesign02_lab_...	dB20(v("/VODIF...				
icDesign02_lab_...	db20Aol	6.01			
icDesign02_lab_...	Aol	1.998			
icDesign02_lab_...	BW	11.13M			
icDesign02_lab_...	GBW	22.02M			
icDesign02_lab_...	UGF	19.7M			

3) Differential and CMFB loops stability (STB analysis):

- Run STB analysis (in addition to AC analysis 1Hz:10Gz, logarithmic, 10 points/decade) two times: first using the 0V source in the diff path, and second using the 0V source in the CM path.
- Plot loop gain in dB and phase vs frequency for the two simulations overlaid.

Common loop gain in yellow and it's phase in red

Differential loop gain in Cyan and it's phase in green



• Compare GBW and PM of diff and CM loops. Comment.

Differential results

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
icDesign02_lab_...	Loop Gain Phase				
icDesign02_lab_...	Loop Gain dB20				
icDesign02_lab_...	Phase Margin	88.89			
icDesign02_lab_...	GBW(Diff)	9.538M			

Common results

Test	Output	Nominal	Spec	Weight	Pass/Fail
Filter	Filter	Filter	Filter	Filter	Filter
icDesign02_lab_...	Loop Gain Phase				
icDesign02_lab_...	Loop Gain dB20				
icDesign02_lab_...	Phase Margin	61.87			
icDesign02_lab_...	GBW(CM)	14.95M			

The results indicate that the differential response will be overdamped with no ringing/peaking and that the CM loop response will be underdamped with slight ringing and overshoot and peaking in frequency domain, also since the bandwidth of the differential loop is higher by 3k we expect it to be faster than the CM loop we need to keep that in mind and make sure that the common loop responds to changes.

• Compare DC LG and GBW of the diff loop with those obtained from open-loop simulation. Comment.

From open loop-simulation we reached a gain of 70dB and if we assume $B=0.5$

We get

$$LG = B * A_{ol} = 64dB$$

$$\text{with feedback } LG = 61dB$$

, the feedback network that we have put loaded the amplifier and B calculations are not precise that's the main reason for this variation in results between OL and with feedback simulations and that's the main reason why we insisted on reaching higher gain in the OL simulations

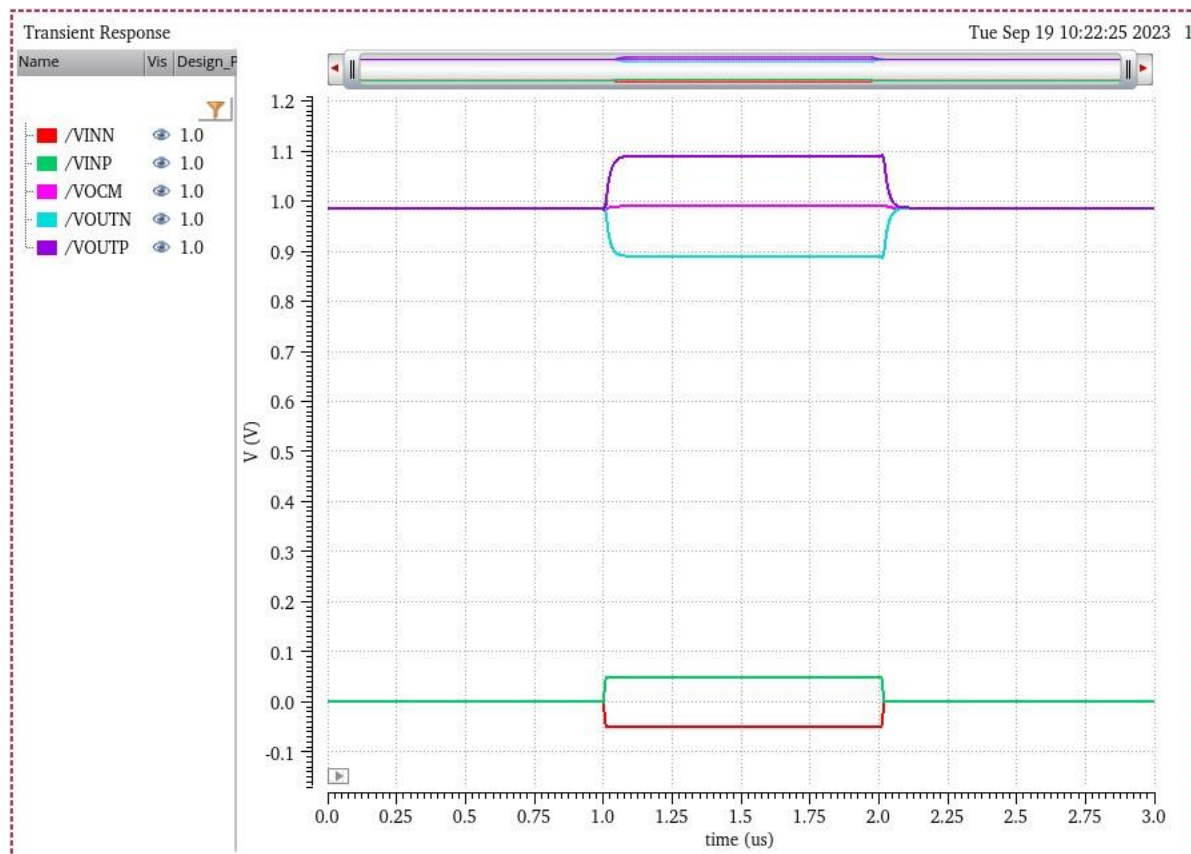
PART 6: Closed Loop Simulation (Transient Analysis)

Use the same testbench as in Part 5, but change diff and CM input sources as explained below.

Report the following:

1) Differential and CMFB loops stability (transient analysis):

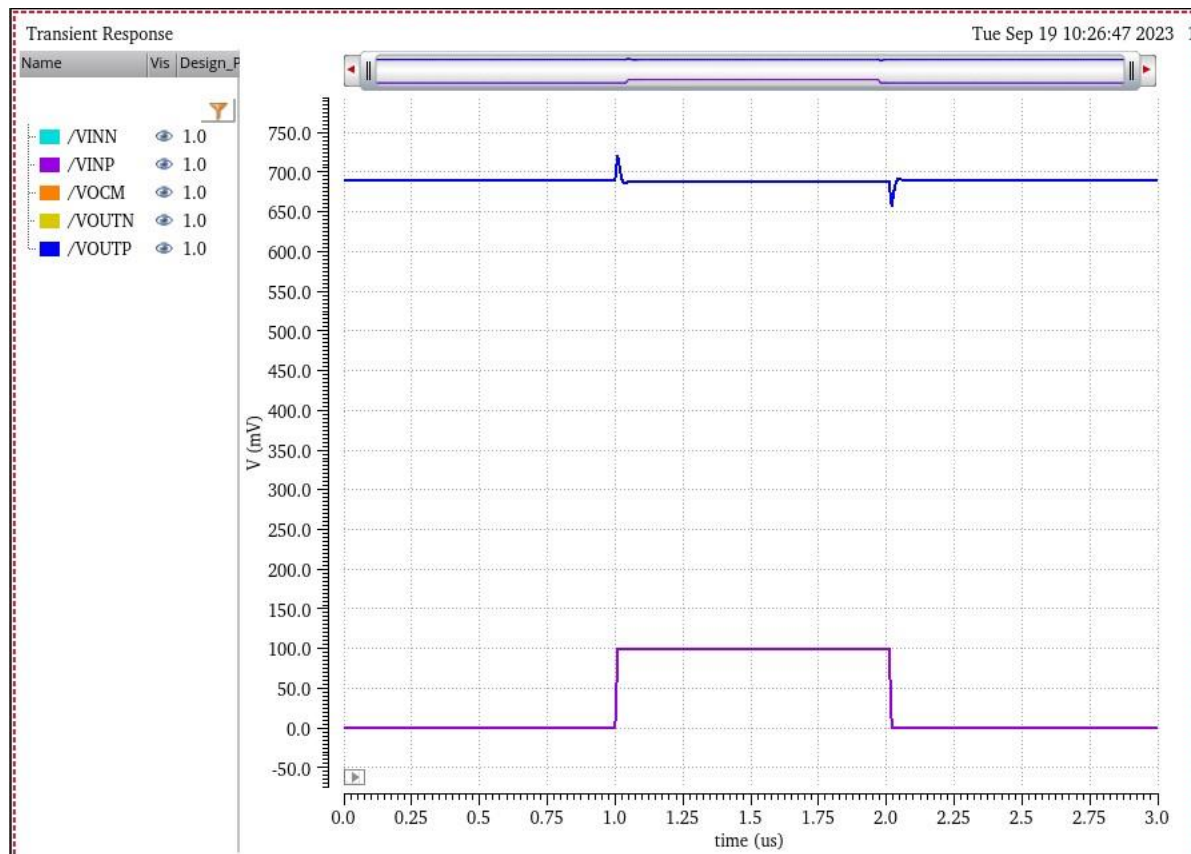
- Apply a differential input pulse (initial value = 0, pulse value = 100mV, delay = 1us, period = 2us, pulse width = 1us, rise = fall = 10ns).
- Run transient analysis for 3us with 10ns max step.
- Plot the transient signals at VINP, VINN, VOUTP, VOUTN, and VOCM overlaid in the same figure.



- Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

there is no noticeable ringing for either, both loops should be stable with adequate PM

- Set differential input to zero and apply the same previous pulse at the balun CM input.
- Run transient analysis for 3us to test the fully differential capacitive amplifier stability.
- Plot the transient signals at VINP, VINN, VOUTP, VOUTN, and VOCM overlaid in the same figure.



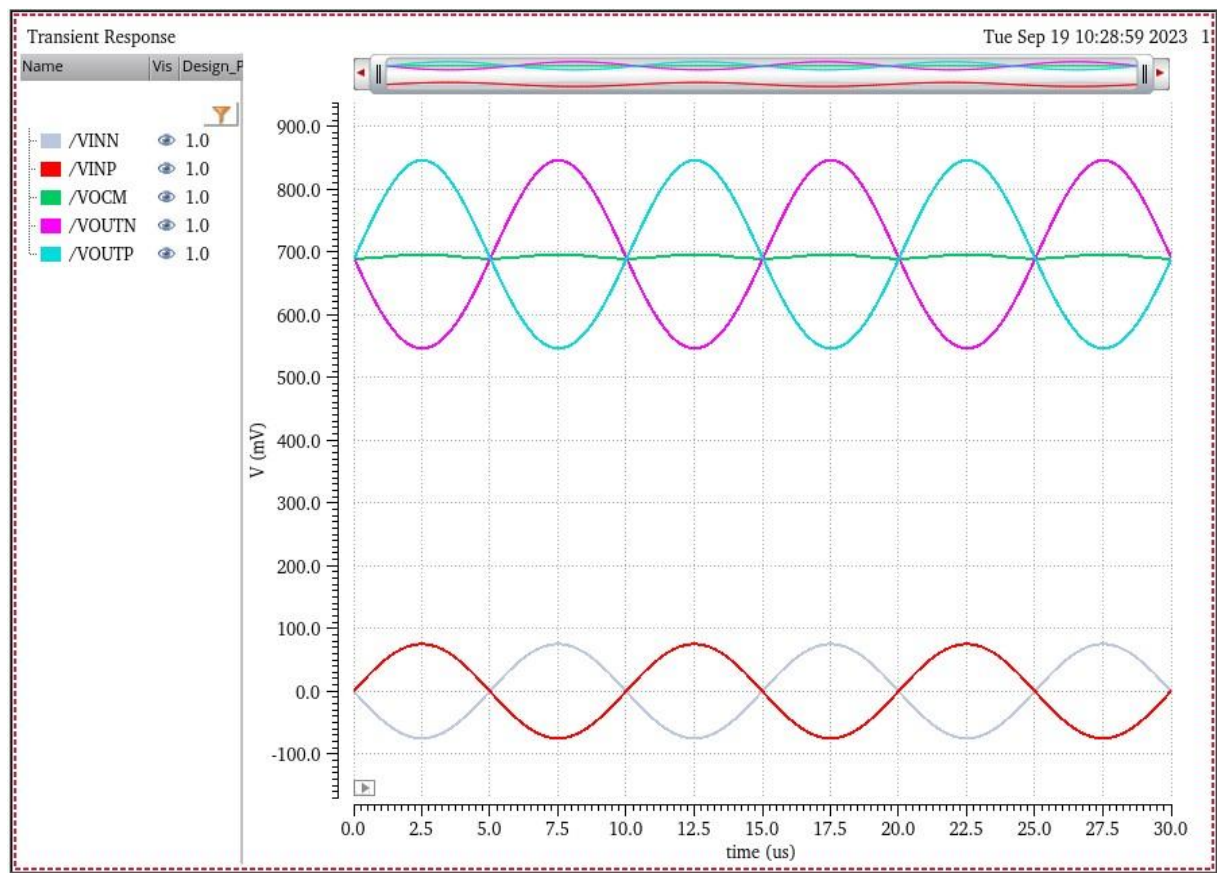
- Do you notice any differential/CM ringing? Are both loops stable with adequate PM?

For differential output there is no ringing but for CM there is very slight overshoot(remember PM is around 62), both loops should be stable with adequate PM

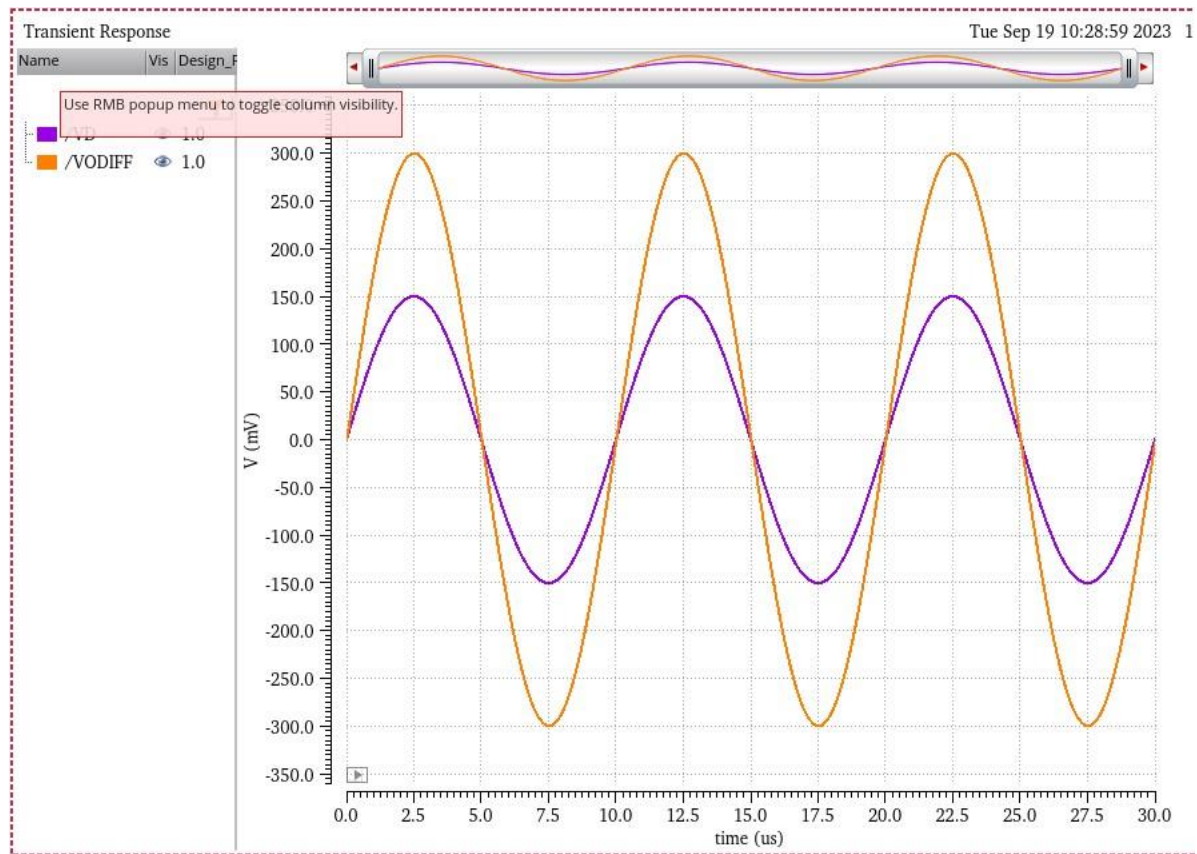
2) Output swing:

- Apply a differential sinusoidal input with freq = 100kHz and amplitude = 150mV.
- Run transient analysis for three periods (30us) with 0.1us max time step.

- Plot the transient signals at VINP, VINN, VOUTP, VOUTN, and VOCM overlaid in the same figure.



- Plot the transient signals at VIDIFF and VODIFF overlaid in the same figure.



- Calculate the diff input and output peak-to-peak swings and the closed loop gain

$$\text{diff input peak} - \text{peak} = 150 - (-150) = 300 \text{ mV}$$

$$\text{diff output peak} - \text{peak} = 300 - (-300) = 600 \text{ mV}$$

$$A_{cl} = \frac{V_{ODFF_{pp}}}{V_{DIFF_{pp}}} = 2$$